DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

Probabilistic Machine Learning

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Probabilistic inference

Marginalisation

Conditioning

$$p(z_j|x_i = \bar{x}_i) = \frac{p(\bar{x}_i, z_j)}{p(\bar{x})} = \frac{\int p(x,z)dx_{-i}dz_{-j}}{\int p(x,z)dx_{-i}dz}$$

Factorisation

$$p(x_1 \ldots, x_n) =$$
$$p(x_2, \ldots, x_n | x_1) p(x_1) =$$
$$p(x_3, \ldots, x_n | x_1, x_2) p(x_2 | x_1) p(x_1) =$$

$$p(x_n | x_1, \ldots x_{n-1}) p(x_{n-1} | x_1, \ldots, x_{n-2}) \ldots p(x_3 | x_2, x_1) p(x_2 | x_1) p(x_1)$$

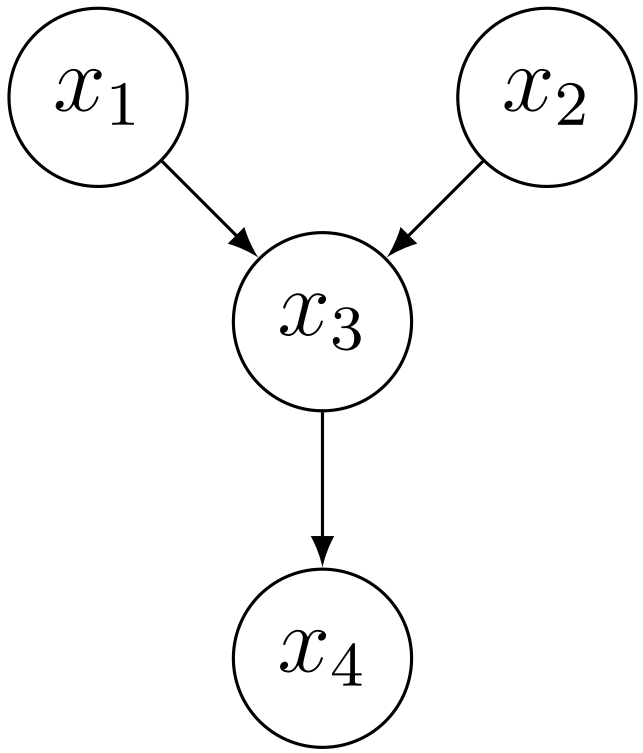
$$p(x_1, \cdots, x_n) = p(x_n | x_{n-1}) p(x_{n-1} | x_{n-2}) \cdots p(x_2 | x_1) p(x_1)$$

Probabilistic Graphical Models

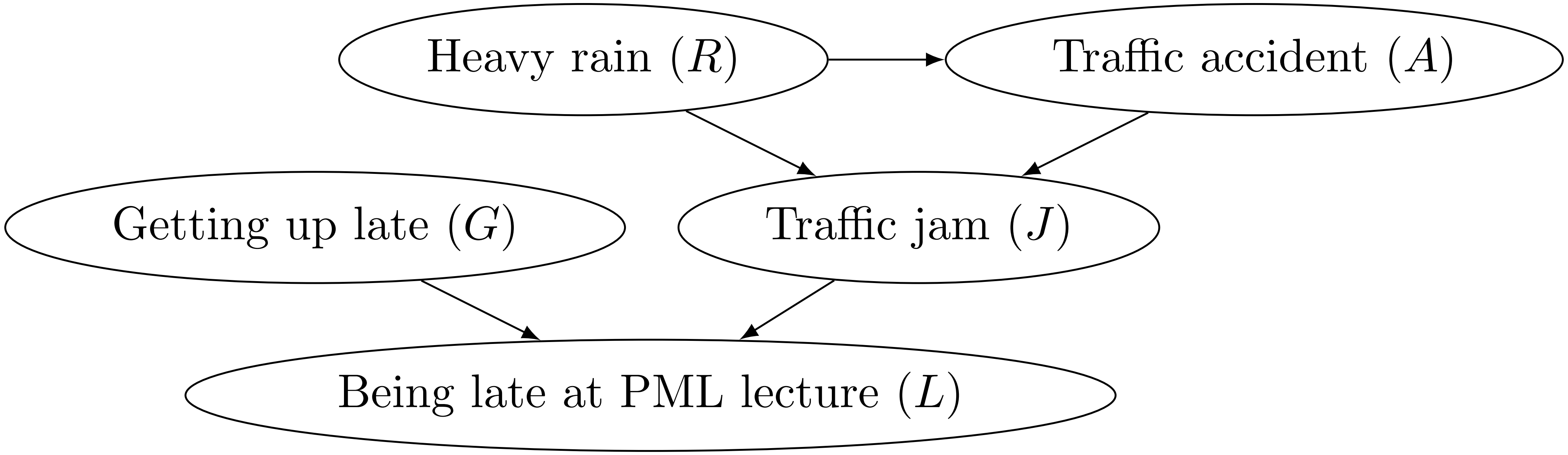
- Bayesian networks
- Markov Random Fields
- Factor Graphs

Bayesian Networks

$$p(x_1, x_2, x_3, x_4) = p(x_4|x_3)p(x_3|x_2, x_1)p(x_2)p(x_1)$$

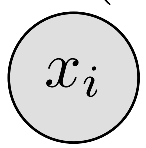


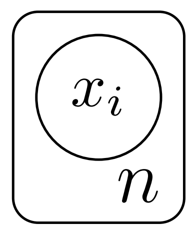
Bayesian Networks



$$p(L, G, J, R, A) = p(L|G, J)p(J|R, A)p(A|R)p(R)p(G)$$

Bayesian Networks

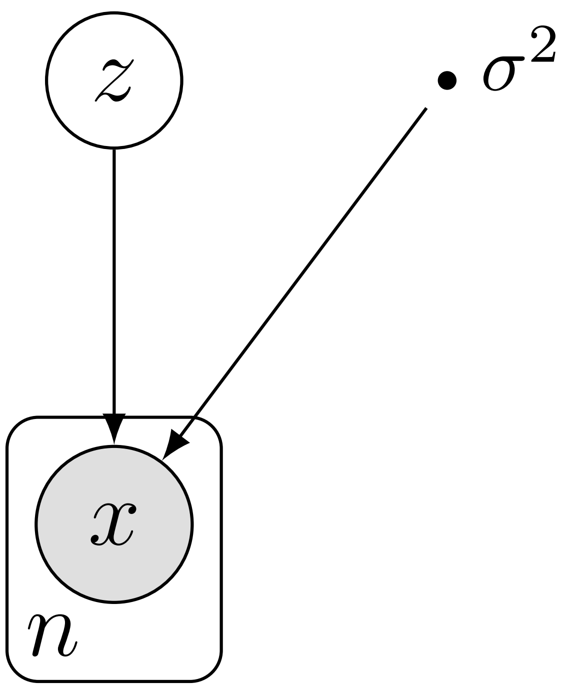
Observed nodes (Random variables of which we know the value) are **coloured** or shadowed 

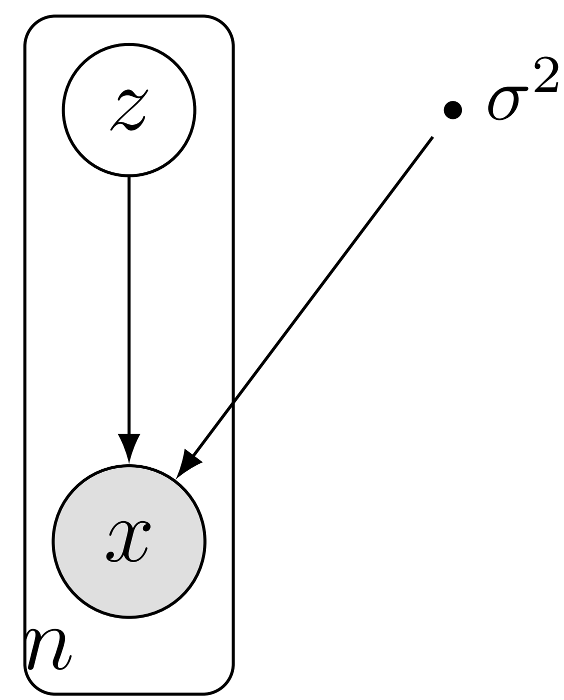
Plated nodes are a shorthand for a collection of nodes. The following represents the nodes $x_1, \ldots, x_n$ 

Deterministic quantities represented as small solid circles. These are fixed parameters of the model (represented graphically as $\bullet\ \alpha$)

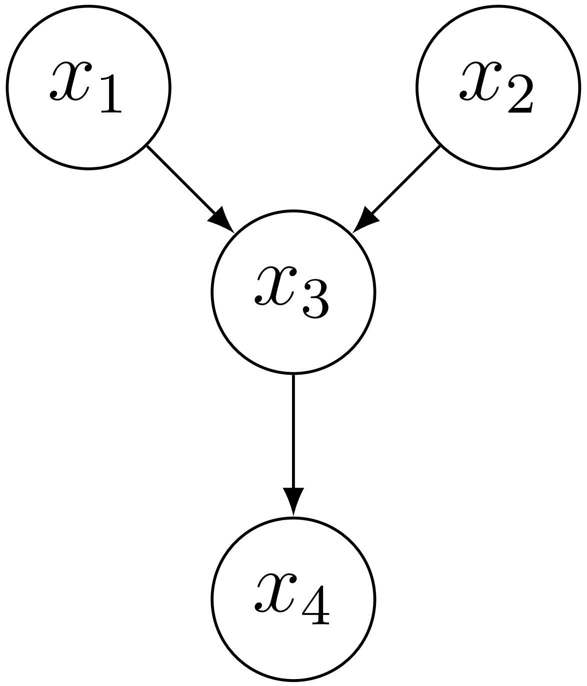
Mixtures of gaussians

$$p(x, z) = p(x|z)p(z)$$

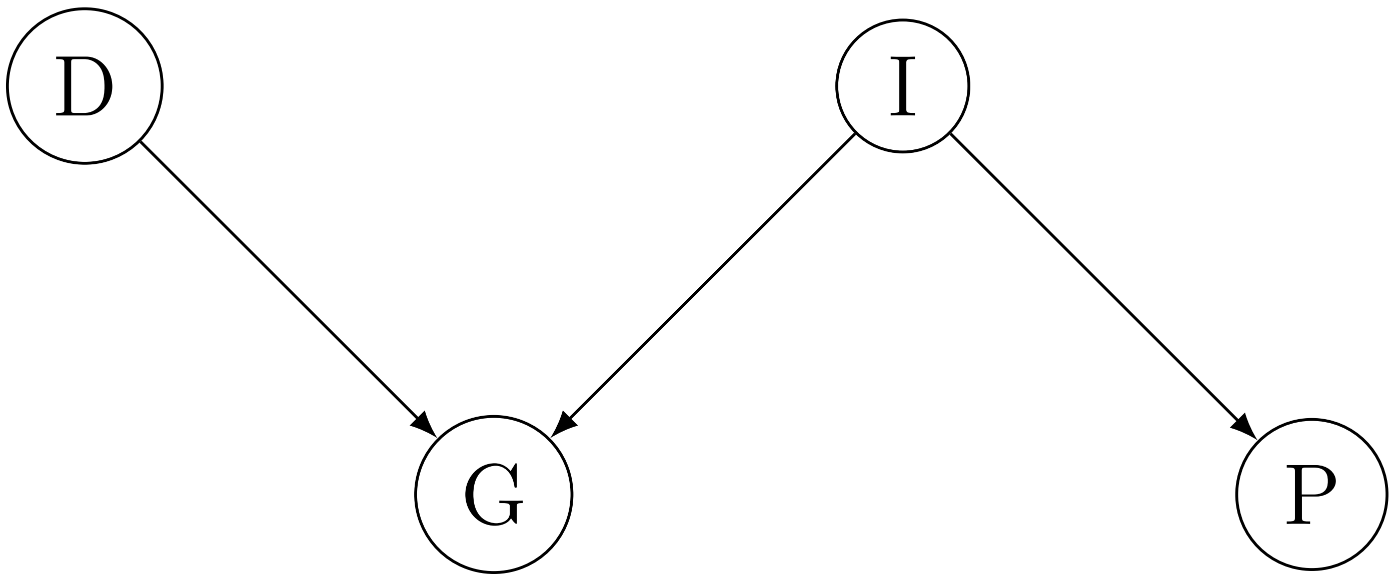




Ancestral sampling

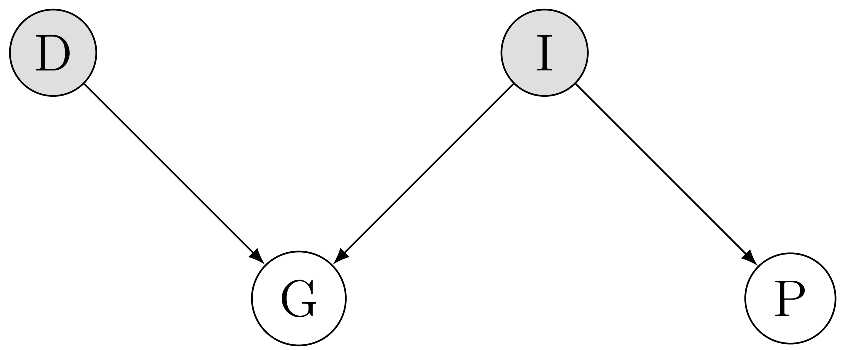


Reasoning

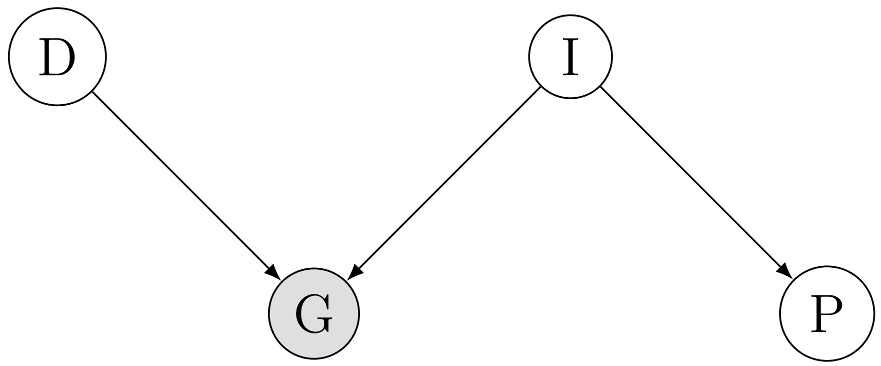


Reasoning

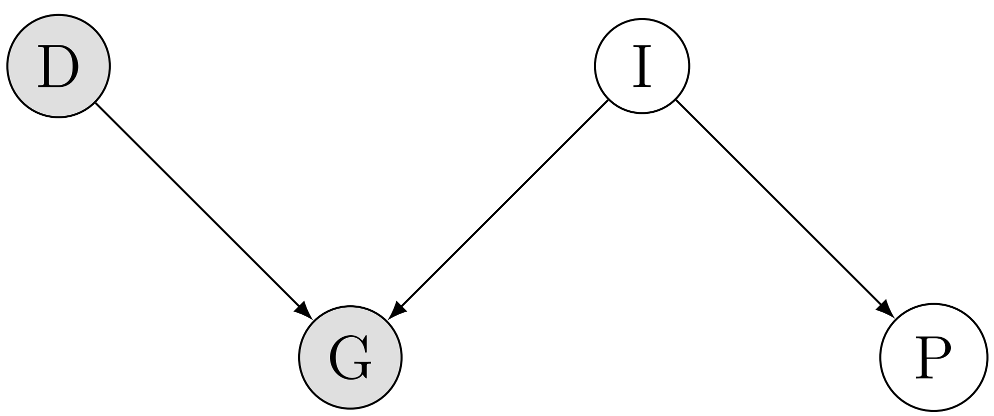
- **Causal Reasoning**



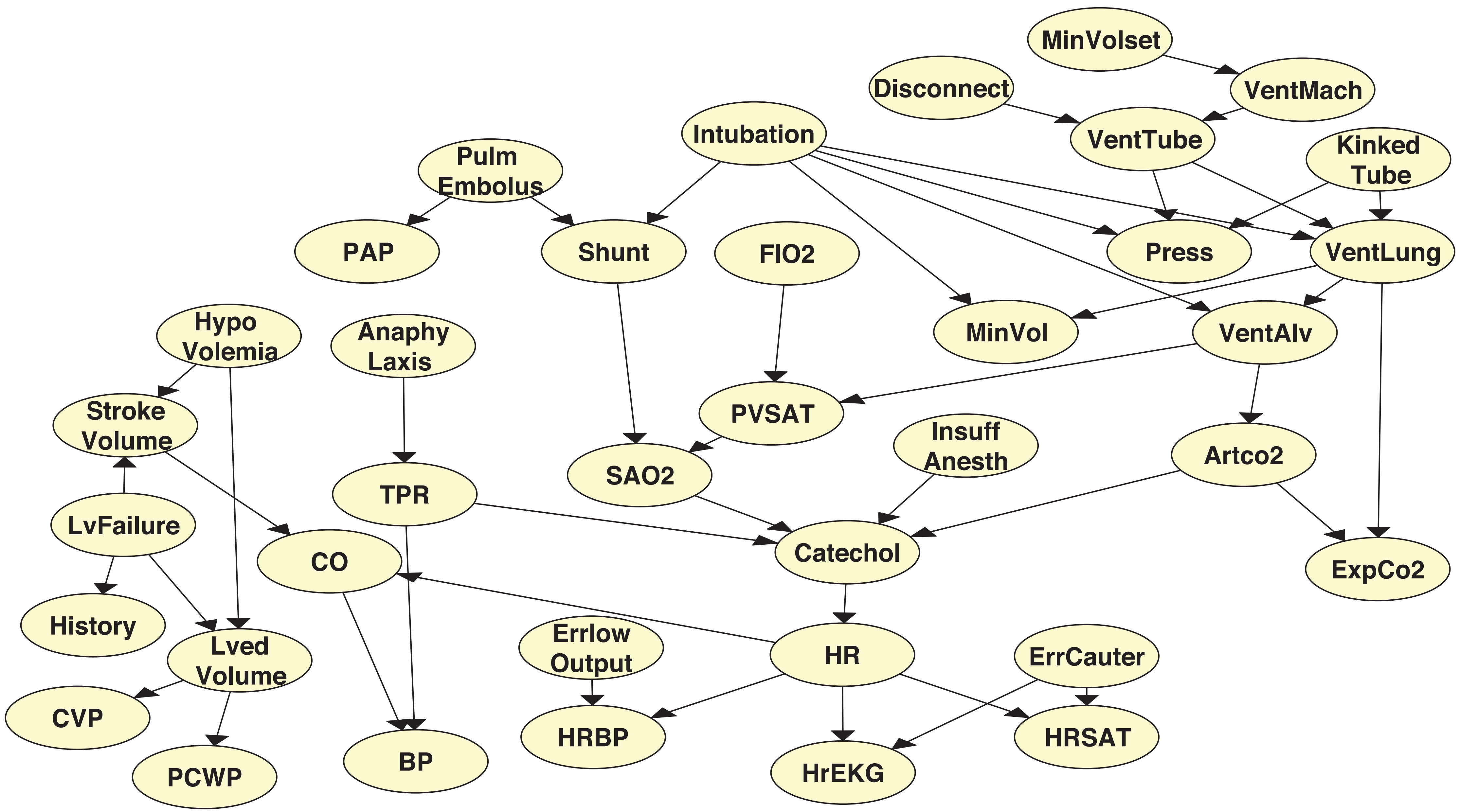
- **Evidential reasoning**



- **Intercausal reasoning**



BN - example

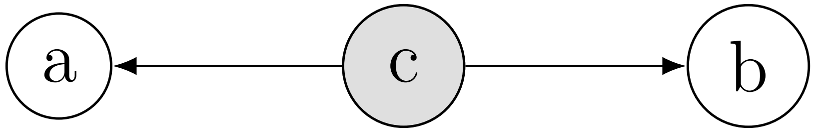


Expert System: Intensive Care Alarms

Conditional Independence

4.1 Tail to tail

Definition 4. *Consider the RVs a, b, c. We say that a and b are tail to tail with c iff* $p(a, b, c) = p(a|c)p(b|c)p(c)$



$$p(a, b) = \sum_c p(a|c)p(b|c)p(c) \neq p(a)p(b)$$

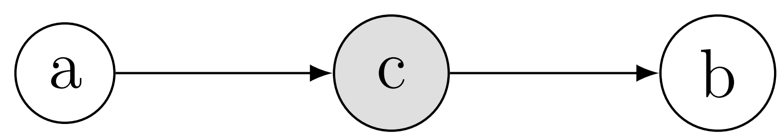
Also

$$p(a, b|c) = \frac{p(a, b, c)}{p(c)} = \frac{p(a|c)p(b|c)p(c)}{p(c)} = p(a|c)p(b|c)$$

Conditional Independence

4.2 Head to tail

Definition 5. *Consider the RVs a, b, c. We say that a and b are head to tail with c iff $p(a,b,c) = p(a)p(c|a)p(b|c)$ or $p(a,b,c) = p(b)p(c|b)p(a|c)$*



$$p(a,b) = \sum_c p(c|a)p(b|c)p(a) = p(b|a)p(a)$$

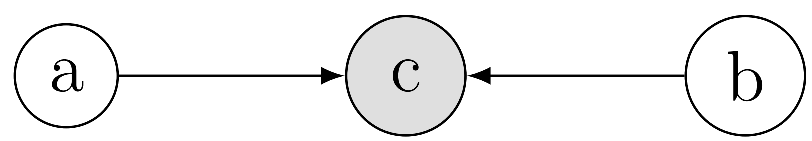
Also

$$p(a,b|c) = \frac{p(b|c)p(c|a)p(a)}{p(c)} = p(b|c)p(a|c)$$

Conditional Independence

4.3 Head to head

Definition 6. *Consider the RVs a, b, c. We say that a and b are head to head with c iff $p(a, b, c) = p(a)p(b)p(c|ab)$*



$$p(a, b) = \sum_c p(c|a, b)p(b)p(a) = p(b)p(a)$$

Also

$$p(a, b|c) = \frac{p(c|a, b)p(b)p(a)}{p(c)} \neq p(b|c)p(a|c)$$

Conditional Independence

Definition 7. *Given RVs a,b,c, a path from a to b is* ***blocked*** *by c iff:*

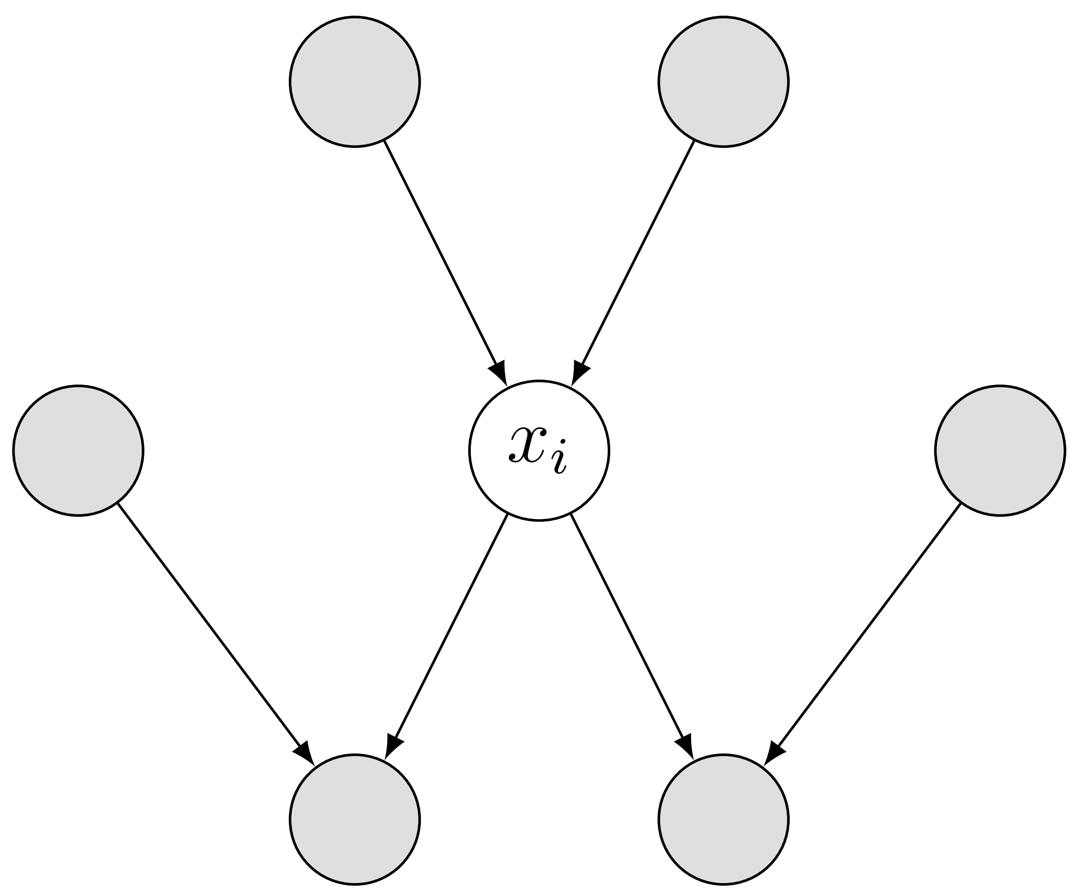
- *c is observed and the path is head-to-tail or tail-to-tail in c.*
- *c is not observed, nor any descendant of c, and the path is head-to-head in c.*

Proposition 1. *Let A, B, C subsets, if all paths from a node in A to a node in B are blocked by a node in C then $A \perp\!\!\!\perp B | C$*

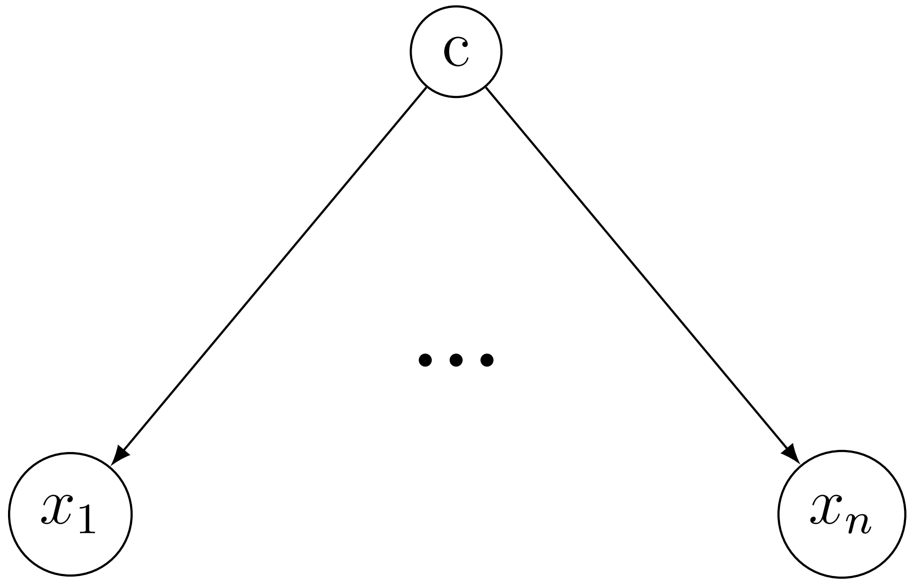
Markov Blanket

$$x_{-i} = \{x_1, ..., x_{i-1}, x_{i+1}, ..., x_n\}.$$

$$p(x_i|x_{-i}) = \frac{p(x_1, \ldots, x_n)}{p(x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n)} = \frac{\prod_j p(x_j|pa_j)}{\sum_{x_i} \prod_j p(x_j|pa_j)}$$



Naive Bayes



$$p(x_1, \ldots, x_n | c) = \prod_{i=1}^{n} p(x_i | c)$$

$$p(c | x_1, \ldots, x_n) \propto p(x_1, \ldots, x_n | c) p(c) = p(c) \prod_{i=1}^{n} p(x_i | c)$$

log odd ratio

$$\frac{p(c = 0 | x_1, \ldots, x_n)}{p(c = 1 | x_1, \ldots, x_n)}$$

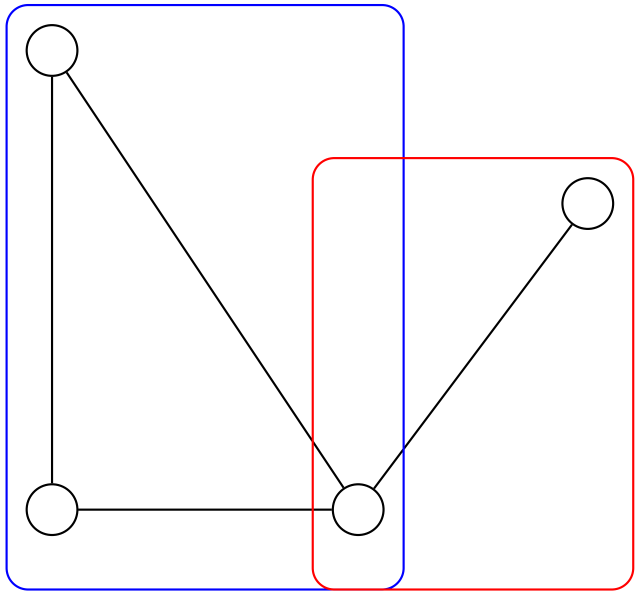
Markov Random Fields

conditional independence

Proposition 2. *Consider three subsets A, B, C.* $A \perp\!\!\!\perp B|C$ *iff all paths from a node* $a \in A$ *to any node* $b \in B$ *pass from C (i.e. are blocked by a node in C).*

Definition 8. *A* ***Markov Blanket*** *in a Markov Random Field for a given* x_i *is the set of neighbours of* x_i

MRF - factorisation



no edge from x_i to x_j

$$p(x_i, x_j | x_{-\{i,j\}}) = p(x_i | x_{-\{i,j\}}) p(x_j | x_{-\{i,j\}})$$

Definition 9. *A **clique** is a fully connected subgraph.*

Definition 10. *A **maximal clique** is a clique that cannot be extended.*

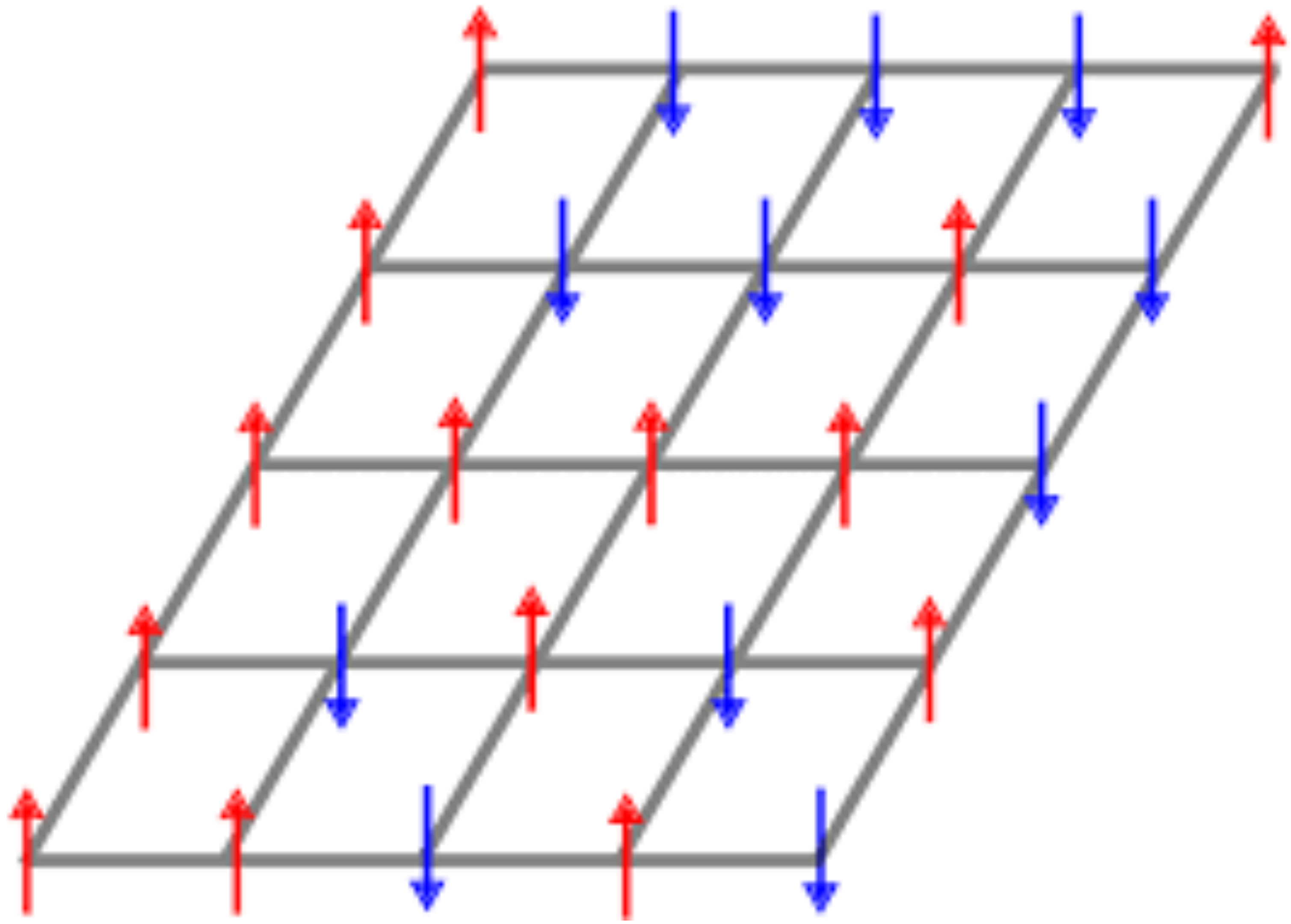
$$p(x) = \frac{1}{Z} \prod_{c \in \mathcal{C}} \Psi_C(x_C)$$

$$Z = \int \prod_C \Psi_C(x_C) dx_C < \infty$$

$$\Psi_C(x_C) = \exp(-E(x_C))$$

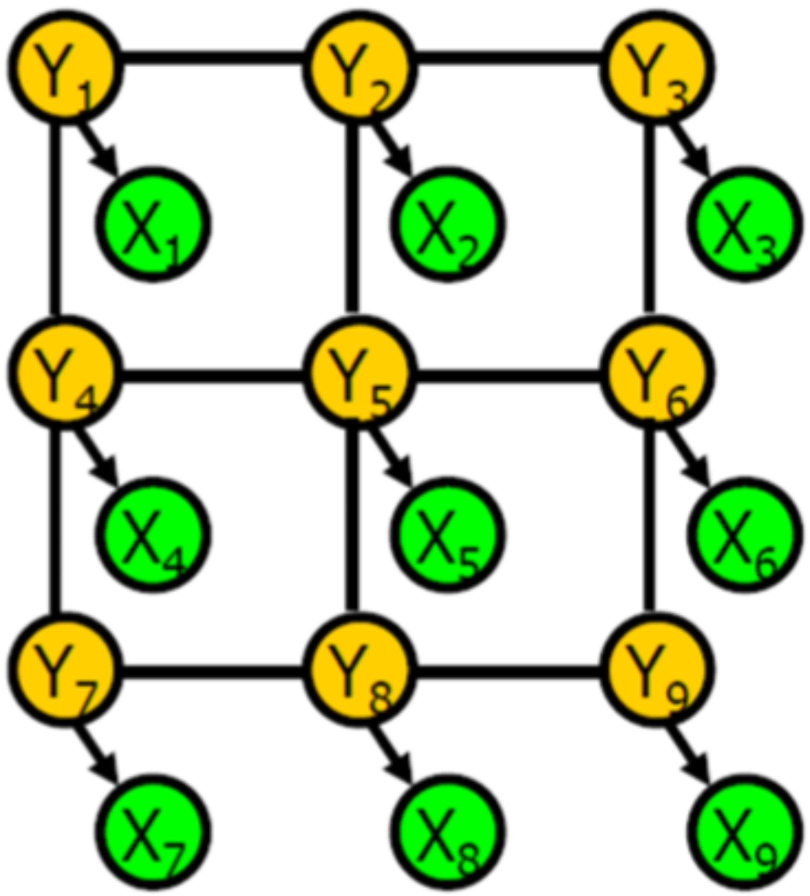
$$\Psi_{C1}(x_{C1}) \Psi_{C2}(x_{C2}) = \exp(-E_1(x_{C1})) \exp(-E_2(x_{C2})) = \exp(-E_1(x_{C1}) - E_2(x_{C2}))$$

MRF - Ising Model



Ising Model (binary RVs, energy: $Jx_1 x_2$, per edge (x_1, x_2))

MRF - image denoising



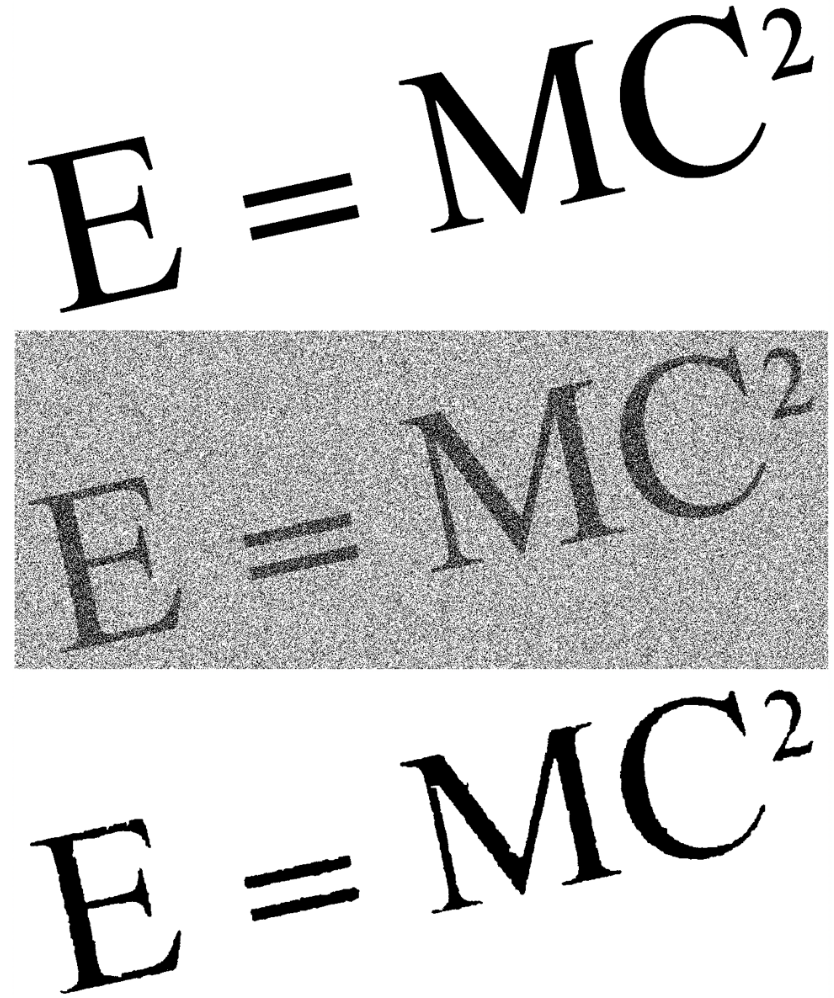
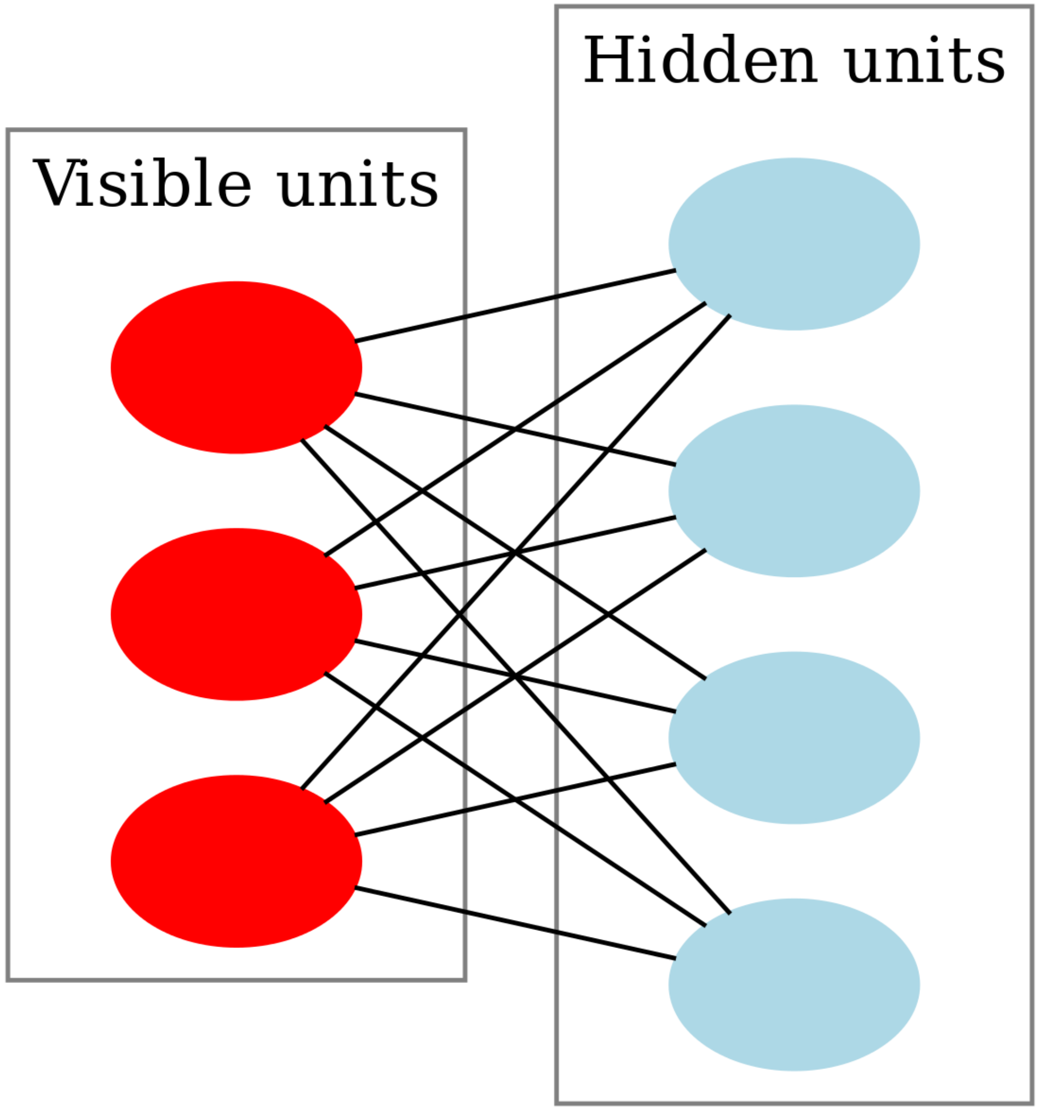


Image denoising: $x_{ij}, y_{ij} \in \{0, 1\}$

MRF - Restricted Boltzmann Machines



Restricted Boltzmann Machine: bipartite graph, binary nodes